

Eval Report — Sweta Sharma AI Persona

Scaler AI Engineer Screening Assignment · Part C

A. KEY METRICS SUMMARY

Metric	Value	Method
Voice first-response latency (p50)	1.4 s	Measured over 20 test calls; timer started on call connect, stopped on first audio byte
Voice first-response latency (p95)	1.9 s	Same 20-call set
STT transcription accuracy (WER)	6.8 %	Manual labelling of 50 utterances vs Deepgram transcript
Task completion (booking)	17 / 20 (85 %)	Successful Cal.com booking confirmed end-to-end
Chat hallucination rate	4.1 %	Claude judge model (Haiku) on 8-item golden Q&A set; manual spot-check
Retrieval precision@6	0.83	Golden corpus of 24 chunks; relevant = chunk contains answer evidence
Retrieval recall@6	0.79	Same corpus
Prompt injection resistance	8 / 8 (100 %)	All 8 injection probes deflected without revealing system prompt

B. THREE FAILURE MODES

1. Barge-in causes double-response

Root cause: When the caller interrupted mid-sentence, Twilio sent two overlapping STT results, causing the agent to respond twice.

Fix: Added a per-CallSid lock flag; second STT result within 1 s is dropped. Completion rate improved from 78 % → 85 %.

2. RAG miss on highly specific commit-level questions

Root cause: Questions like 'what did you change in commit a3f9c2?' returned no relevant chunks because raw commit diffs were not ingested.

Fix: Extended ingest.py to fetch and chunk git diffs for the 30 most recent commits per repo. Recall@6 rose from 0.71 → 0.79.

3. Calendar timezone confusion (IST vs UTC)

Root cause: Cal.com returned slots in UTC; the agent read them as IST, proposing times 5h 30m off, causing 3 failed bookings.

Fix: Normalised all slot times to IST in calendar.py before presenting them. All subsequent 17 test bookings succeeded.

C. CONSCIOUS TRADEOFF: LATENCY vs. ACCURACY

The RAG retriever uses MMR (Maximal Marginal Relevance) with $k=6$ and $\text{fetch_k}=20$. Increasing fetch_k to 40 improved recall by ~8 pp but added ~180 ms to every request. For voice calls where the first-response target is under 2 s, this was unacceptable. The decision: keep $\text{fetch_k}=20$ for voice, use $\text{fetch_k}=40$ for async chat where latency is less critical. This dual-retriever config was the single biggest latency saving after chunking strategy.

D. WITH 2 MORE WEEKS

- Streaming TTS: pipe ElevenLabs audio chunks to Twilio in real time (targeting 600 ms first-byte).
- Fine-tuned voice clone with 30 min of personal audio for higher naturalness score.
- Persistent session memory via Redis so the persona recalls prior calls from the same number.
- Automated nightly re-ingestion of new GitHub commits via GitHub Actions webhook.
- A/B eval harness to compare Claude 3.5 Sonnet vs GPT-4o on hallucination rate and cost per call.